Konstantinos Konstantinidis

Ph.D., Iowa State University

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Skills

Proficient: Python • C++ • AWS • MATLAB • PyTorch • MPI • MapReduce
Good: Java • SQL • Hadoop • NumPy • Bash • Git • Jenkins

Education

Iowa State University — Ames, IA
Ph.D. & M.Eng., Electrical and Computer Engineering, GPA: 3.94/4, Advisor: Prof. Aditya Ramamoorthy. — *Dec. 2022*

Technical University of Crete — Chania, Greece
Diploma (5-year program), Electrical and Computer Engineering, GPA: 3.4/4, Advisor: Prof. George Karystinos. — *Dec. 2016*

Industry Experience

Senior Research Scientist at Meta — Menlo Park, CA | 12/2023 – Present
I work on the Marketplace Recommendation & Delivery Infrastructure building highly-scalable ML pipelines for feature extraction/serving/logging, item ranking and training data generation, primarily in C++ and Python. Drove LLM-assisted Offline-to-Online conversion workstream of 6+ engineers to holistically monitor and fix offline-to-online topline metric conversion.

- Independently landed multiple capacity efficiency wins: ∼**$4,828,000/year** capex savings, **73%** latency reduction, **2,500+** host savings, **5.9**× compilation speedup, **32.9PB** of data warehouse storage.
- Developed *AutoScaling* solution achieving **97.7%** load-testing health and **4.5** → **0.1** lagging incidents/month.
- Built DAG framework for RecSys training data generation supporting realtime model prediction, feature fetching and request optimization supporting **14+** usecases including CTR, CVR, 3rd-party products (eBay), and in-feed recommendations.
- Developed ML feature infra for the following usecases: user sequential features, media ranking, privacy compliance, multi-model support for realtime user embeddings.
- Implemented realtime *detection* for lagging, coverage, and quality issues of features across Marketplace & Dating; work recognized at Facebook level. Led *Oncall Efficiency* across **3** teams, bringing false-positive-rate from **27%** → **10%**.

Software Engineer, Platform at C3 AI — Redwood City, CA | 09/2022 – 11/2023
As a member of the Platform - Data team, I developed C3 AI's *feature store* system that stores and serves machine learning features. My work involved alleviating bottlenecks identified through algorithmic analysis and profiling (e.g., *YourKit*) across the data pipeline, hence improving the latency of feature serving and enabling the system to scale distributively across multiple nodes.

Software Engineer Intern at Meta — Menlo Park, CA | 05/2022 – 08/2022
Developed debugging tools for machine learning feature authoring used in Facebook Marketplace. The implemented framework categorizes errors during feature compilation, generates alerts, and assigns tasks to the appropriate team; I integrated it with the CI/CD. Another end product of my work was a UI tool to retrieve and transform feature values from low-latency storage.

Software Engineering Intern, Platform at C3 AI — Redwood City, CA | 06/2021 – 08/2021
Implemented a framework for cluster failure prediction; its first component is the data pipeline which loads cluster health metrics, handles missing data, and creates a training data set. The second component is the ML pipeline which trains a model and predicts the cluster's state using streaming data. Followed the process of continuous integration / continuous deployment (CI/CD).

Research Projects

ByzShield: Robust distributed learning | ***Python, PyTorch, AWS, MPI, Bash.*** — 01/2020 – 04/2021
Developed a defense for distributed learning in which computing devices may return erroneous or malicious gradients. The method achieves a **20**% increase in top-1 accuracy on the **CIFAR-10** dataset and a **36**% reduction in the fraction of corrupted gradients.

CAMR: Aggregated MapReduce over multiple jobs | ***Python, AWS, MPI, Bash.*** — 12/2018 – 04/2020
Proposed a method to reduce the MapReduce communication overhead for aggregate functions. It achieves state-of-the-art communication load but requires an exponentially smaller number of jobs and achieved speedup **4.3**× over the baseline approach.

Straggler mitigation in matrix multiplication | ***Python, MPI, AWS.*** — 01/2018 – 01/2019
Proposed and implemented a technique to tolerate the presence of servers that suffer from slow computation. Our algorithm can alleviate a higher number of slow servers and requires **60**% of the time needed by prior methods on actual AWS EC2 simulations.

MapReduce communication load reduction | ***C++, AWS, MPI, Hadoop, HDFS, Bash.*** — 05/2017 – 12/2018
Proposed an algorithm to reduce **MapReduce** communication load; it uses *Single Parity Check* codes and design theory to assign tasks to servers and splits files less finely than prior work. Tweaked the **TeraSort** algorithm (to sort data sets in **HDFS**). The method uses **MPI** for communication. It supersedes the state-of-the-art by **2.6**× and the baseline approach by **4.7**× on AWS EC2 clusters.

Personal Project

Options Trading Bot | ***Python, Schwabdev, yfinance, Bash.*** 11/2025 – Present

Developed options trading library with real-time market data from **Schwab API** and **Yahoo Finance API**. Features: option analysis, portfolio updates, and *Telegram* bot for control.

- Implemented algorithms for volatility-based strike selection, delta/gamma risk assessment, and *knapsack* algorithm for optimal combination discovery under profit constraint.
- Engineered remote API interaction with retry, backoff, and handling of stale data ("999" values).
- Employed *Fernet* encryption for sensitive Schwab tokens, **Google Firestore** for key storage.
- Developed automated *Google Sheets* updates, including concurrent edit detection/resolution.

Awards

- **Best Student Poster Award**, Midwest Machine Learning Symposium (MMLS), June 2019 (link).
- **Research Excellence Award**, Iowa State University, May 2022.
- **Teaching Excellence Award**, Iowa State University, May 2019.
- **Graduate Scholarship**, Gerondelis Foundation, Lynn, MA, June 2018.

Publications (Google Scholar)

- **K. Konstantinidis**, N. Vaswani, and A. Ramamoorthy, "Detection and Mitigation of Byzantine Attacks in Distributed Training," *IEEE/ACM Transactions on Networking (ToN)*, October 2023.
- **K. Konstantinidis** and A. Ramamoorthy, "Aspis: Robust Detection for Distributed Learning," *IEEE International Symposium on Information Theory (ISIT)*, June 2022.
- **K. Konstantinidis** and A. Ramamoorthy, "ByzShield: An Efficient and Robust System for Distributed Training," *Machine Learning and Systems (MLSys)*, April 2021.
- **K. Konstantinidis** and A. Ramamoorthy, "Resolvable Designs for Speeding up Distributed Computing," *IEEE/ACM Transactions on Networking (ToN)*, August 2020.
- **K. Konstantinidis** and A. Ramamoorthy, "CAMR: Coded Aggregated MapReduce," *IEEE International Symposium on Information Theory (ISIT)*, July 2019.
- L. Tang, **K. Konstantinidis** and A. Ramamoorthy, "Erasure coding for distributed matrix multiplication for matrices with bounded entries," *IEEE Communication Letters (COMML)*, November 2018.
- **K. Konstantinidis** and A. Ramamoorthy, "Leveraging Coding Techniques for Speeding up Distributed Computing," *IEEE Global Communications Conference (GLOBECOM)*, December 2018.